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Comparative assessment of volatiles in juices and essential oils from minor *Citrus* fruits (Rutaceae)

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Abstract

Citrus Essential Oils (CEOs) from the peel of six minor *Citrus* fruits (*C. bergamia* 'Castagnaro,' 'Fantastico,' and 'Femminello,' *C. × myrtifolia*, *C. mitis*, and *C. japonica*) were obtained by hydrodistillation and cold hand pressing of the flavedo to investigate their volatile constituents and antimycotic activities. The volatile compounds present in the juices of the same fruits, obtained by manual squeezing of the endocarps, were evaluated by direct immersion and Solid Phase Micro Extraction. Monoterpenes were identified as the main class of components in all the six CEOs. Limonene was the most abundant compound, while BEOs (Bergamot Essential Oils) changed its composition according to the different extraction method and the fruit varieties. Non-metric Multidimensional Scaling (NMDS) of the matrix taxa x chemical compounds showed that five chemical compounds had a significant discriminative function among the samples: high amounts of limonene characterized the cluster of *C. japonica*, *C. × myrtifolia*, and *C. mitis*, while the group of three cultivars of *C. bergamia* differed in its predominance of linalool and linalyl acetate. Furfural and 4-terpineol were identified in the highest amounts in the *Citrus* juices analyzed by direct immersion. BEO obtained by hydrodistillation of *C. bergamia* 'Fantastico' was the most active in the antifungal tests. The antimycotic activity of *C. japonica* EOs against dermatophytic species was also demonstrated. Based on our results, the use of these products would appear an interesting alternative in producing topical herbal products for the treatment of affected animals or for the purpose of environmental fungal control.

KEYWORDS

Bergamot, calamondin, chinotto, cluster analysis, kumquat, non-metric multidimensional scaling

1 | INTRODUCTION

The Rutaceae family is represented by more than 300 species, widely distributed and cultivated worldwide. The main genus is represented by *Citrus* plants, the most economically important food crops in the family, but also important as garden ornamentals.

They are trees or flowering evergreen plants that generally have a strong scent with fruits, such as the modified sectional berry known as a hesperidium and recognized for their fragrance. *Citrus* fruits are widely cultivated and consumed throughout the world; they have a pleasant smell and taste and are used as a food, even though there are some differences in flavor among citrus species or varieties.¹

Abbreviations: BEOs, Bergamot Essential Oils; CEOs, *Citrus* Essential Oils; EO, Essential Oil; NMDS, Non-metric MultiDimensional Scaling.

Citrus essential oils (CEOs) are 'Generally Recognized As Safe' (GRAS) and widely used in foods, beverages, liqueurs, or as food flavoring agents but also as aromatic agents in soaps, perfumes, and household products. CEOs have shown antimicrobial and antifungal activities together with antioxidant effects. CEOs inhibit a wide range of Gram-positive and Gram-negative bacteria, both in direct oil and vapor form.² Monoterpenes predominate in CEO composition with limonene as the major chemical component, ranging from 32% to 98%. Many authors ascribe the antifungal effect of CEOs to the presence of volatile compounds, mainly monoterpenes and sesquiterpenes.³

Citrus bergamia Risso (bergamot) is a hybrid between bitter orange (*C. aurantium* L.) and lemon (*C. limon* L. Burm.) or a mutation of the latter. It is produced almost exclusively in the Reggio Calabria area (Calabria region, South Italy). However, although in small quantities, it also grows in other countries such as Greece, Morocco, Iran, the Ivory Coast, and in a few South American countries.⁴ Bergamot is mostly used to produce EO, obtained by wash-scraping the peel of the fruits. It used to be extracted by hand, but today it is extracted by a mechanical cold-pressing procedure (peel machine). It is widely used in pharmaceutical industries for its antiseptic and antibacterial properties, in cosmetic industries for its intense fragrance and freshness, and in food industries as an aroma in the preparation of sweets, liquors, and tea. Currently, three different cultivars are cultivated in Calabria: 'Femminello,' 'Castagnaro,' and, mainly, 'Fantastico' which is characterized by a good yield and essential oil quality representing about 75% of the whole production.⁵ Bergamot essential oil (BEO) contains several bioactive molecules particularly effective as an antimicrobial for both human and plant pathogens. It is mainly constituted of monoterpene and sesquiterpene hydrocarbons including monoterpene limonene as the main compound, and high quantities of oxygenated compounds, such as linalool, linalyl acetate, γ -terpinene, and β -pinene.⁶ The bergamot juice, in contrast to those from other *Citrus* fruits, is considered a waste product of the essential oil production; until now, use in the food industries has not been found because of its bitter taste. Recently, HPLC and LC-MS analyses of the juice have identified different phenolic compounds such as ascorbic acid, flavanones such as neohesperidin, naringin, and neoeriocitrin and lesser amounts of flavones and furanocoumarins such as bergapten and bergamottin.⁷ All these compounds give high antioxidant properties to bergamot juice, which therefore represents a good resource of antioxidants and ascorbic acid and may contribute to an increase in the healthy properties of *Citrus* juices.⁸ Regarding the volatile composition of the two bergamot cultivars 'Femminello' and 'Fantastico' cultivated in the Calabria region of Italy, Marzocchi and collaborators⁹ observed that the aroma profile of the cold pressed bergamot oil was affected by the ripening stage with significant variation in the amount of monoterpene and sesquiterpenes. The cultivar 'Fantastico' was considered the most aromatic bergamot useful for industrial application.⁹

Citrus × myrtifolia (Ker Gawl.) Raf. (in Italy called 'Chinotto') was previously classified as *Citrus aurantium* L. var. *myrtifolia* Guill., but today The Plant List Database [http://www.theplantlist.org/

browse/A/Rutaceae/Citrus/] reports these names as synonyms of *Citrus × aurantium* L. This minor *Citrus* has indeed been classified by genetic markers as a mutant of *Citrus × aurantium* L.¹⁰ Its fruits are like a small tangerine, have a characteristic scent and green when unripe changing to a deep orange and a bitter-sour taste when ripe. A very popular commercial carbonated soft drink named 'Chinotto,' which has a bitter taste and digestive properties, is made with the juice of the fruits of this plant. Locally, *Citrus × myrtifolia* is considered a niche product and used in the Italian confectionery industry to prepare jams and candies, while its essential oil is preferred for use in perfumery. Few papers are present in the scientific literature regarding the bioactive compounds in *Citrus × myrtifolia* fruits. 'Chinotto' juice is rich in organic acids, sugars and flavonoids, especially naringin and neohesperidin, and phenolic acids together with carotenoids.¹¹ The juice from immature fruits showed to be richer in bioactive compounds than that obtained from ripe fruits.^{12,13} The essential oil composition is also reported and limonene (80.1%), linalool (5.5%), linalyl acetate (0.8%), nootkatone (0.7%), and farnesol (0.5%) were identified as the main constituents in fruits collected in Finale Ligure (Liguria, Italy).¹⁴

Calamondin or calamansi, commonly named 'naranjo chino,' is known as *Citrus mitis* Blanco. It is a hybrid of *Citrus reticulata* Blanco and *C. japonica* Thunb. The fruits are small, ovate, edible, and orange-reddish in color with few seeds. Calamondin is widely distributed in tropical and subtropical areas of the world and it is the leading *Citrus* fruit in the Philippines, where it is primarily used for juice and as a substitute for lemon. The fruit is eaten together with the peel and used as a brewed fruit tea.¹⁵ Many flavonoids are present in the calamondin peel which increases after ripening. The flavonoids found in the calamondin fruit were diosmin, hesperetin, naringin quercetin, and hesperidin.¹⁶ The volatile fragrance of the calamondin juice was determined by headspace analysis (SPME), and monoterpene aldehydes, alcohols, and hydrocarbons were mainly identified.¹⁷ However, the most prominent compound in the peel oil was limonene (77.0%), while elemol (15.4%) and α -eudesmol (11.1%) represented the main components in the leaf oil.¹⁸ More recently, the EOs from calamondin flowers and leaves were analyzed: monoterpene hydrocarbons dominated in the flower oil with limonene (74.5%) as the major component, while sesquiterpene (82.0%) with elemol was the major component (37.5%) characterized in the leaf oil.¹⁹

Kumquat (*Citrus japonica* Thunb., synonyms *Fortunella margarita* (Lour.) Swingle or *Fortunella japonica* (Thunb.) Swingle) is the smallest of the true *Citrus* fruits native to China. The small round-oval fruits (no longer than 2 cm in diameter) are orange yellow. The raw fruit is eaten without removing the peel and is both sweet and sour.¹⁸ It is added to fruit salads or used for decorative purposes and is suitable for various products such as candied fruit, wine, marmalade, and fruit tea.²⁰ Kumquat was used for a long time in traditional herbal medicine, especially for colds and coughs due to its antiphlogistic, carminative, and expectorant properties. It was also used as a dietary supplement since it is an excellent source of nutrients and phytochemicals, including ascorbic acid, carotenoids, flavonoids,

and essential oils. This latter evidenced limonene as the principal component (93.8%) together with myrcene (2.7%) and δ -germacrene (1.34%).²¹ In a recent study performed on kumquat peel extracted by different methods (ie hydrodistillation, solvent and supercritical carbon dioxide), a similar composition was evidenced with limonene, myrcene and germacrene D as the main constituents but in different percentages.²²

The principal aim of this work was to evaluate the chemical composition of CEOs from the peel of the following minor *Citrus*: three cultivars of *C. bergamia* Risso ('Castagnaro,' 'Fantastico,' and 'Femminello'), together with *C. × myrtifolia* (Ker Gawl.) Raf., *C. mitis* Blanco and *C. japonica* Thunb. CEOs were obtained by two different methods: hydrodistillation and cold hand pressing of the flavedo. The volatiles of these fruit juices, obtained by manual squeezing of the endocarps, were also evaluated by Direct Immersion-SPME (solid-phase microextraction). Since the aroma composition of CEOs and fruit juices are very important for their use in different industries, their analysis becomes important for their exploitation and destination of use. The antifungal activity of the hydrodistilled CEOs was also tested to explore the potential benefits of these plant extracts as a source of naturally occurring bioactive agents to be included in different end products, such as herbal products, functional food, dietary supplements, cosmetics, or medicines.

2 | EXPERIMENTAL

2.1 | Plant material and sample preparation

The bergamot fruits were directly supplied by 'Visalli Diego Srl.' (Melito di Porto di Salvo, Reggio Calabria, Italy) and identified by one of the authors, who is a botanist (DC). *Citrus × myrtifolia* plants (PI034267) with fruits were provided by 'Pamparino Green & Parking s.a.s.' nursery (Finale Ligure, Savona, Italy), while the plants of *C. mitis* (PI034265) and *C. japonica* (PI034266) were purchased from a local nursery (Green Service, Pisa, Italy).

The fresh *Citrus* fruits were peeled, and the flavedos (3-5 fruits for each sample) were hydrodistilled in a Clevenger apparatus (Tecnovetro, Milan, Italy) at 100°C for 2 hours. After distillation, the obtained EOs were put into vials and stored at -4°C until analysis. The EO yields are reported in Table 1. A small part of the fresh *Citrus* peel (only the flavedo from one fruit for each sample, about 1-2 g according to the fruit size) was manually cold pressed to recover the essential oil from the rind in a glass vial. A very low amount of the EO was recovered from each fruit, but enough for our investigation, then diluted with *n*-hexane before the GC-MS analysis. The amount collected from each manually cold pressed EO was about 3-5 droplets. After peeling off the rind, the remaining endocarp of each fresh *Citrus* fruit was squeezed to obtain the juice. The juices obtained from each fruit were filtered with cellulose paper (Whatman® Qualitative Filter Paper, Merck KGaA) and kept at -4°C until analysis.

2.2 | Direct immersion-solid phase MicroExtraction (DI-SPME) for volatile analysis in *Citrus* juice

This technique has been adopted to evaluate the volatiles present in the *Citrus* juices previously obtained after separately squeezing the endocarp of all the fruits with a manual juicer. Each juice sample was filtered with cellulose paper (Whatman® Qualitative Filter Paper, Merck KGaA) before analysis (3 mL). The DI technique was performed with Supelco SPME devices coated with DVB/CAR/PDMS 50/30 μ m, Stableflex 24Ga, (Supelco®) to evaluate the volatiles inside the juice. The fiber was maintained inside each sample for the necessary time, according to the analyzed material (usually 30 seconds, experimentally determined to avoid any deterioration of the fiber). After the sampling time, the fiber was withdrawn into the needle and then immediately transferred to the injection port of GC-MS.

2.3 | Identification of volatile components by GC-MS

The GC/MS analyses of the *Citrus* essential oils were carried out with a Varian CP-3800 gas chromatograph (Varian Inc, Palo Alto) equipped with a DB-5 capillary column (30 m $\times$ 0.25 mm i.d., film thickness 0.25 μ m) and a Varian Saturn 2000 ion-trap mass detector (Varian Inc, Palo Alto). The oven temperature was programmed, rising from 60 to 240°C at 3°C/min; injector T° 220°C; transfer-line T° 240°C; carrier gas He (1 mL/min); injection volume 0.2 μ L (10% hexane); split ratio 1:30.

The constituent identification was performed by comparing the retention time (t_R) with that of authentic samples, comparing their linear retention indices (LRI) relative to a series of *n*-hydrocarbons, and on computer matching against a commercial library^{23,24} and also by the use of a homemade library of mass spectra built up from pure substances and components of known EOs as well as MS literature data.

2.4 | Statistical analysis

All data from the three different methods of extraction were merged in a global matrix taxa $\times$ chemical compounds (values under 1.5% were excluded for the purpose of statistical analysis). We performed a cluster analysis (CA) using average-linkage clustering and the Bray-Curtis similarity index (proper to matrix with biological data, characterized by the presence of many zeros) for a square root transformed matrix (18 taxa per 35 compounds). The groups obtained with the CA were analyzed with the Good Clustering Level (GCL) method, analyzing the variability among and inside the groups. The same resemblance matrix was used to perform non-metric multidimensional scaling (NMDS), a technique that represents samples in a low-dimensional space by optimizing the correspondence between the original dissimilarities and the

TABLE 1 Compounds identified in the selected *Citrus* essential oils (BEOs and CEOs) obtained by hydrodistillation

N°	Compounds	Retention index		Citrus samples					
				BEOs			CEOs		
				<i>C bergamia</i> 'Castagnaro'	<i>C bergamia</i> 'Fantastico'	<i>C bergamia</i> 'Femminello'	<i>C x myrtifolia</i>	<i>C mitis</i>	<i>C japonica</i>
				Relative content (%)					
1	α -thujene	932	930	0.3	0.1	0.1	- ^a	-	-
2	tricyclene	934	927	1.1	0.4	0.5	0.8	0.5	0.6
3	sabinene	978	975	0.9	0.5	0.6	0.3	0.4	0.2
4	β -pinene	981	979	5.6	3.2	3.5	2.8	1.2	tr ^b
5	myrcene	993	991	1.5	1.8	1.5	2.7	1.9	2.7
6	<i>n</i> -octanal	999	999	-	tr	tr	0.3	0.3	-
7	α -terpinene	1019	1017	0.2	0.2	0.2	tr	-	-
8	<i>o</i> -cymene	1024	1026	0.2	0.4	0.2	tr	-	-
9	limonene	1032	1029	41.8	25.0	21.4	84.1	89.2	92.7
10	(Z)- β -ocimene	1042	1037	0.3	0.5	0.4	tr	-	-
11	(E)- β -ocimene	1053	1050	0.7	1.0	0.9	0.4	tr	tr
12	γ -terpinene	1062	1060	7.0	6.0	5.9	0.1	tr	tr
13	<i>n</i> -octanol	1068	1068	-	-	-	-	0.1	-
14	<i>trans</i> -linalool oxide (furanoid)	1073	1074	-	tr	0.1	0.4	-	-
15	terpinolene	1089	1089	0.6	0.7	0.7	-	tr	-
16	linalool	1102	1097	16.7	34.3	28.1	2.0	0.5	0.6
17	<i>n</i> -nonanal	1104	1101	-	-	-	tr	0.4	-
18	citronellal	1153	1153	-	-	tr	-	0.1	-
19	borneol	1165	1169	-	0.1	-	-	-	-
20	<i>n</i> -nonanol	1174	1170	-	-	tr	-	0.1	-
21	4-terpineol	1180	1177	0.2	0.3	0.5	0.4	tr	tr
22	α -terpineol	1192	1189	2.5	5.2	5.5	1.1	0.2	tr
23	methyl chavicol	1198	1196	-	-	-	-	0.1	-
24	<i>n</i> -decanal	1206	1202	-	tr	-	0.2	0.5	-
25	octanol acetate	1214	1214	-	-	0.2	0.1	0.2	-
26	<i>trans</i> -carveol	1221	1217	-	-	-	0.3	-	-
27	nerol	1228	1230	-	0.8	1.4	-	-	-
28	neral	1242	1238	-	0.2	0.4	-	-	-
29	carvone	1248	1243	-	-	tr	0.4	tr	tr
30	(E)-2-decenal	1266	1264	--	-	-	-	0.2	-
31	linalyl acetate	1260	1257	17.1	13.7	20.0	0.5	-	-
32	geranial	1267	1267	-	0.2	0.5	-	-	-
33	perilla aldehyde	1271	1272	-	-	-	tr	-	0.1
34	(E,E)-2,4-decadienal	1318	1317	-	-	-	-	0.2	-
35	α -terpinyl acetate	1352	1349	0.2	0.2	0.4	tr	-	-
36	neryl acetate	1368	1362	0.7	1.5	2.1	0.1	0.1	-
37	geranyl acetate	1386	1381	0.8	2.3	3.2	0.3	0.8	0.3
38	β -caryophyllene	1418	1419	0.4	0.4	0.4	tr	-	tr
39	α - <i>trans</i> -bergamotene	1437	1435	0.4	0.3	0.3	-	-	-
40	γ -muurolene	1477	1480	-	tr	-	tr	0.8	-

(Continues)

TABLE 1 (Continued)

N°	Compounds	Retention index		Citrus samples					
				BEOs			CEOs		
				<i>C bergamia</i> 'Castagnaro'	<i>C bergamia</i> 'Fantastico'	<i>C bergamia</i> 'Femminello'	<i>C x myrtifolia</i>	<i>C mitis</i>	<i>C japonica</i>
				Relative content (%)					
41	germacrene D	1481	1485	-	-	tr	-	-	2.6
42	bicyclgermacrene	1500	1500	-	tr	tr	-	tr	0.3
43	β -bisabolene	1509	1506	0.4	0.4	0.5	-	-	-
44	germacrene B	1561	1561	-	-	-	-	tr	0.1
45	(E)-nerolidol	1566	1563	-	-	tr	0.1	-	-
46	β -eudesmol	1651	1651	-	-	-	-	0.6	-
47	α -eudesmol	1654	1654	-	-	-	-	0.4	tr
48	<i>n</i> -eicosane	2000	2000	-	-	-	0.6	-	-
49	<i>n</i> -tricosane	2300	2300	-	-	-	0.1	-	-
50	<i>n</i> -pentacosane	2500	2500	-	-	-	0.8	-	-
Chemical groups (%)									
	Monoterpene hydrocarbons			60.1	39.7	35.9	91.3	93.2	96.1
	Oxygenated monoterpenes			38.1	58.7	62.1	5.5	1.7	1.0
	Total monoterpenes			98.3	98.4	98.0	96.8	94.9	97.0
	Sesquiterpene hydrocarbons			1.2	1.0	1.2	-	0.8	2.9
	Oxygenated sesquiterpenes			-	-	-	0.1	1.0	-
	Total sesquiterpenes			-	-	-	0.1	1.8	2.9
	Phenylpropanoids			-	-	-	-	0.1	-
	Not terpenic compounds			-	-	0.2	2.1	2.0	-
	Total identified (%)			99.4	99.4	99.4	100.0	99.9	100.0
	N. of identified compounds			22	31	34	31	29	18
	EO Yield (% w/w)			1.22	0.86	0.97	0.25	<0.01	0.14

Abbreviations: LRI_{exp}, experimental linear retention index on DB-5 column; LRI_{lit}, Linear retention index recovered from the literature.²³

^anot detected;

^btr, trace.

distances in the ordination.²⁵ Pearson correlation has been calculated to analyze the effects of chemical compounds on taxa. The statistical multivariate analyses were done with PRIMER v.7 software (PRIMER-E, Plymouth²⁶).

2.5 | Antimycotic tests

The in vitro antimycotic activity of hydrodistilled CEOs from *C bergamia* ('Fantastico,' 'Castagnaro,' and 'Femminello'), *C x myrtifolia*, and *C japonica* were evaluated on *Saprolegnia parasitica* isolated from water in a rainbow trout farm and kept in Malt

extract broth (MEB). *Saprolegnia* was isolated from the environment, being the less invasive procedure to avoid handling of infected fish which would exacerbate both the clinical picture and infection outcome. Due to the very low amount of calamondin EO (<0.01% w/w), this oil was not tested. The effectiveness of CEOs was assessed by means of a microdilution test carried out in MEB. Portions of approximately 1 mm² mycelium were used as fungal inocula in 96-wells plates (Pbi International). A one-line seed was put in each well to enhance fungal growth, as previously described.²⁷ Stock solutions at 10% of all the chemically defined CEOs were diluted into MEB to obtain concentrations ranging from 0.1% to 10. Positive control consisted of malachite

green ($1 \mu\text{g mL}^{-1}$) and negative controls were provided to evaluate the fungal growth in medium only. The plates were incubated at room temperature for about 5 days until control wells showed a mycotic growth.

The in vitro antimycotic activity of *Citrus bergamia* ('Fantastico,' 'Castagnaro,' and 'Femminello'), *C × myrtifolia*, and *C japonica* was assayed as previously described.²⁸ In detail, dermatophytes from clinical isolates of *Microsporum canis*, *Tricophyton mentagrophytes*, and *Microsporum gypseum*, were employed. The dermatophytes were cultured by pets showing clinical signs of ringworm, during the routine diagnostic activity. In detail *M canis* and *T mentagrophytes* were isolated from two symptomatic cats and *M gypseum* was obtained from a dog. These fungi were isolated by brushing the fur of the suspected animals and seeding the brush directly on a dermatophyte selective medium (Mycobiotic agar, Difco). The inocula consisted of about 1 mm 2 log phase unsporulated dermatophyte mycelium. A stock solution of 20% was achieved in tube, twofold dilutions were serially prepared apart in a microtiter plate, then 200 μL of solution were dispensed in each well, where fungal inocula were added. The fungal isolates were maintained onto Sabouraud dextrose agar (SDA). The antimycotic activity of EOs was assayed by a microdilution test carried out as recommended by the Clinical and Laboratory Standards Institute M38A2 (Clinical and Laboratory Standards Institute [CLSI]²⁹). Stock solutions at 20% of all the chemically defined CEOs were diluted in sweet almond oil to obtain concentrations ranging from 0.01% to 10%. Positive controls using a conventional antimycotic drug (Itraconazole 32 mg/mL) were achieved. Negative controls were provided to evaluate the fungal growth in medium only. The plates were maintained at 28°C, until a mycotic growth was noticed in negative control wells (about one week). MICs for all testing were measured by examining the wells under a stereoscopic microscope.

3 | RESULTS AND DISCUSSION

3.1 | Hydrodistilled Citrus EO

The compositions of the essential oils and the yields obtained by hydrodistillation are reported in Table 1. The most productive citrus fruit was *C bergamia* 'Castagnaro' with 1.22% of EO yield, as were all the BEOs in general, in comparison with the other minor fruits. In total, 50 compounds were identified with a percentage of identification ranging between 99.4% and 100.0%. In all the six considered *Citrus* Essential Oils (CEOs), the monoterpenic class of components represented almost the total of the identified compounds, with percentages ranging among 94.9% and 98.3%. Oxygenated monoterpenes prevailed in the 'Fantastico' and 'Femminello' cultivars of *C bergamia* with percentages of 58.7% and 62.1% respectively, while *C bergamia* 'Castagnaro' was characterized by monoterpene hydrocarbons (60.1%) as the main constituents. The CEOs of the other three minor species were characterized by hydrogenated monoterpenes (*C japonica* > *C mitis* > *C × myrtifolia* with 96.1% > 93.2% > 91.3%,

respectively). Among the main compounds, limonene was present in all the samples of CEOs with the highest amount in the three minor *Citrus* (*C japonica* 92.7% > *C mitis* 89.2% > *C × myrtifolia* 84.1%). Bergamot essential oils (BEOs) showed limonene as the main compound only in *C bergamia* 'Castagnaro' followed by *C bergamia* 'Fantastico' > *C bergamia* 'Femminello' where the principal constituent was represented by linalool (34.3% > 28.1%, respectively). Linalyl acetate was the third compound with very good amounts of BEOs (*C bergamia* 'Femminello,' 20.0% > *C bergamia* 'Castagnaro,' 17.1% > *C bergamia* 'Fantastico,' 13.7%). High percentages (over 5%) of γ -terpinene (in all the three varieties), α -terpineol (in *C bergamia* 'Femminello,' 5.5% > *C bergamia* 'Fantastico,' 5.2%) and β -pinene (only in *C bergamia* 'Castagnaro,' 5.6%) contributed to characterize these BEOs. These results agree with the EO composition from bergamot fruit peels collected in Tunisia, studied by Bouzouita et al (2010) especially for *C bergamia* 'Castagnaro' where the amount of monoterpene hydrocarbons and oxygenated monoterpenes showed similar percentages, while in the other two varieties these percentages were inverted. It is important to note that the amount of sesquiterpenes (around 1.0%-1.2%) found in the Tunisian peels, with α -trans-bergamotene (0.3%) and β -bisabolene (0.4%) as the main compounds, is very similar to that found in our samples.³⁰

The EO composition of *C × myrtifolia*, *C mitis*, and *C japonica* also agrees with the data from the literature for each fruit, where limonene represented the main compounds in the volatile bouquet and with the highest amount of total monoterpene hydrocarbons.^{16,19,23,31}

3.2 | Manual cold pressed Citrus EOs

The hand cold-pressing method of flavedo commonly generated lower yields of essential oils than those obtained by hydrodistillation and were not estimated in this study. The compositions of the EOs obtained by hand cold pressing of the peel of different *Citrus* fruits are reported in Table 2. In total, 43 compounds were identified with a percentage of identification ranging from 93.0% to 99.9%, even though the number of identified compounds was very similar in all the BEOs tested ($n. 21 < 24 < 27$, respectively) with a maximum of 29 in *C japonica*, and a minimum (only 2) in calamondin. The essential oils obtained were characterized by the presence of total monoterpenes with percentages varying between 81.1% and 98.1%. Oxygenated monoterpenes prevailed in the BEOs, *C bergamia* 'Femminello' > *C bergamia* 'Castagnaro' > *C bergamia* 'Fantastico' (83.6% > 72.7% > 55.6%, respectively) even though with relatively low percentages. On the contrary, the other three minor CEOs were characterized by high amounts of monoterpene hydrocarbons, 79.5% in *C japonica*, 84.3% in *C mitis*, and 97.1% in *C × myrtifolia*.

Limonene was present as one of the main compounds in all samples with different percentages, but it characterized the smaller fruit CEOs (*C × myrtifolia* > *C mitis* > *C japonica*, 95.6% > 75.2% > 71.6%, respectively) more than the BEOs. In the last two samples, a significant presence of sesquiterpene hydrocarbons was evidenced (7.4% and 15.1%, respectively), while they showed lower amounts in the

BEOs (*C bergamia* 'Castagnaro' > *C bergamia* 'Fantastico' > *C bergamia* 'Femminello' with 3.4 > 2.2 > 1.7%, respectively). The BEOs differed from each other by high percentages of linalyl acetate and linalool (ranging from 42.7% to 64.6% in the first and from 11.6% to 17.5% in the second). γ -terpinene, another minor monoterpene hydrocarbon, was present with a percentage up to 5% only in *C bergamia* 'Fantastico' (6.8%) followed by β -pinene (4.3%). Only two compounds were identified in the manual cold pressed EO of *C x myrtifolia*, where limonene showed the highest amount (95.6%) followed by 1.5% of myrcene.

It is interesting to note that in general the oxygenated monoterpene percentage was lower in the cold pressed BEOs than in the hydrodistilled oils, except for *C bergamia* 'Fantastico' where the content was stable after each method (55.6% vs 58.7%, respectively). The same was evidenced in the cold pressed CEOs, stable only for the 'Chinotto' EO (97.1% vs 96.8%, respectively).

Comparing the compositions of the hydrodistilled CEOs with those obtained by manual cold pressing, some variations are visible in the oils obtained from the different cultivars of *C bergamia*, but not in the other examined species where a slightly lower amount of limonene was evidenced in the cold pressed CEOs. In particular, in the *C bergamia* 'Castagnaro' and 'Femminello,' heating, due to hydrodistillation, led to a reduction in the oxygenated monoterpene content in favor of those hydrocarbons (from 72.7% to 38.1% for *C bergamia* 'Castagnaro' and from 83.6% to 62.1% in *C bergamia* 'Femminello'), even though the total monoterpene percentages remained unchanged (all > 93%). This variation was also evident in the evolution of the different BEO compositions; BEOs obtained by manual cold pressure were rich in linalyl acetate (ranging between 42.7% and 64.6%) in comparison with the oils recovered after hydrodistillation, where the percentages decreased, ranging from 17.1% to 20.0%. A consequent lower linalool amount was also evidenced (from 11.6% to 17.5%, especially in *C bergamia* 'Fantastico' and 'Femminello,' respectively). This means that heating causes a division of the terpenic ester and linalyl acetate, with a consequent increase in linalool.

The analysis of the cold pressed BEOs showed a difference in the quantitative composition from the BEO obtained from Turkish bergamot by the same procedure³²; the main compounds were limonene > linalyl acetate > linalool, while linalyl acetate was predominant in all three varieties studied herein. The comparison with another cold pressed bergamot essential oil from 'Consorzio del Bergamotto di Reggio Calabria' (in Southern Italy) evidenced a similar composition, but with less linalyl acetate (about half the amount).³³ A similar composition was evidenced in comparison with the Greek cold pressed BEOs, where high percentages of linalyl acetate > limonene > linalool were evidenced together with γ -terpinene which supported the good quality of these BEOs since linalyl acetate is responsible for the characteristic odor of the bergamot fruits.³⁴ Also, the ratio of linalool to linalyl acetate, one of the quality indices of BEO, was 0.27 in all the varieties of the present experiment, as reported in the paper of Masayoshi et al which analyzed the cold pressed oil produced with a Pelatrice-type extractor, directly in Reggio Calabria, Italy (0.29).³⁵

Only two works are present in the literature concerning the chemical composition of the *C japonica* EO obtained by hand cold pressing of the rind.^{21,36} Comparing the previous studies with the current analysis, it is possible to declare that the composition of the two different samples from the literature showed a high percentage of limonene (93.7% and 95.0%, respectively) and myrcene (1.84% and 2.0%), while the sample analyzed here differed due to its low amount of limonene (71.6%) and a higher percentage of germacrene D (12.4%). As reported in other *Citrus* species,³⁶ limonene, despite being generally a major proportion of peel oil, had no importance in the flavor impact of their characteristic aroma.

The Calamondin EO obtained by cold pressing was also different to that studied in Taipei, Taiwan (*Citrus microcarpa* Bunge) where monoterpenes dominated with respect to the EO composition obtained by steam distillation; in our sample the percentage of this class of compounds was reversed. However, the sesquiterpenes found in the hand cold pressed EO (7.4%) were higher than those evidenced in the hydrodistilled EO (1.8%), probably because of the non-distillation method which avoids heat stress.³⁷ The composition of the *C x myrtifolia* hand cold pressed EOs was reported here for the first time since no literature has been available until now on this topic.

3.3 | Juice volatile evaluation by direct immersion-solid-phase MicroExtraction (DI-SPME)

The volatile chemical compounds identified in the *Citrus* juices are reported in Table 3. The method of extraction was by direct immersion of the DVB/CAR/PDMS fiber inside the juice and then directly inserted into the GC-MS. In total, 72 compounds were identified with a percentage of identification ranging from 86.0% to 99.9%, varying from the high numbers of identified compounds found in *C bergamia* 'Castagnaro' and 'Femminello' (38) to those identified in *C japonica* juice (only 9). Monoterpene compounds were the main class of components in all the analyzed samples, with a range of identification varying from 90.4% to 44.4% in the following order: *C x myrtifolia* > *C bergamia* 'Fantastico' > *C bergamia* 'Castagnaro' > *C bergamia* 'Femminello' > *C japonica* > *C mitis*, even though in this latter juice (calamondin) a good amount of non-terpenic compounds was also evidenced. Monoterpene hydrocarbons (MH) predominated in three samples (with percentage up to 74.3%), except in *C bergamia* 'Castagnaro,' where oxygenated monoterpenes (OM) showed the highest amount (51.6%). In *C bergamia* 'Femminello,' a similar percentage of the two monoterpene classes of volatiles was evidenced (MO 36.9% and OM 34.5%). Limonene was the main component present in all the analyzed juices in the following order: *C x myrtifolia* > *C japonica* > *C bergamia* 'Fantastico' > *C bergamia* 'Femminello' > *C mitis* > *C bergamia* 'Castagnaro,' with percentages that range from 26.5% to 77.0%. Only two components were present in all the analyzed juices in addition to limonene, myrcene (ranging from 0.7% to 2.6%) and furfural (from 0.3% to 11.6%, this latter in *C mitis*).

Only a few studies are present in the literature regarding the *Citrus* juice volatilome and the qualitative-quantitative analysis was mostly

TABLE 2 Compounds identified in all the *Citrus* essential oils (BEOs and CEOs) obtained by manual cold pressing

N°	Compounds	Retention index		Citrus samples					
				BEOs			CEOs		
				<i>C bergamia</i> 'Castagnaro'	<i>C bergamia</i> 'Fantastico'	<i>C bergamia</i> 'Femminello'	<i>C x myrtifolia</i>	<i>C mitis</i>	<i>C japonica</i>
		LRI _{exp}	LRI _{lit}	Relative content (%)					
1	α -thujene	932	930	tr ^a	0.2	tr	tr ^b	-	-
2	tricyclene	934	927	0.4	0.7	0.3	-	1.8	1.8
3	camphene	955	954	tr	tr	tr	-	-	0.8
4	sabinene	978	975	0.4	0.8	0.2	-	tr	0.2
5	β -pinene	981	979	2.0	4.3	1.0	-	1.9	0.7
6	myrcene	993	991	0.4	0.7	0.3	1.5	2.2	1.9
7	α -phellandrene	1006	1003	-	tr	-	-	1.8	0.2
8	δ -3-carene	1011	1008	-	-	-	-	1.5	1.2
9	α -terpinene	1019	1017	-	0.2	-	-	-	0.2
10	<i>p</i> -cymene	1026	1025	-	-	-	-	-	0.3
11	limonene	1032	1029	15.1	28.4	6.3	95.6	75.2	71.6
12	1,8-cineole	1036	1031	-	-	-	-	-	0.7
13	(<i>Z</i>)- β -ocimene	1042	1037	-	-	-	-	tr	0.4
14	(<i>E</i>)- β -ocimene	1053	1050	0.1	0.2	tr	-	-	0.1
15	γ -terpinene	1062	1060	4.2	6.8	1.8	-	-	0.2
16	terpinolene	1089	1089	0.2	0.4	0.1	-	-	-
17	linalool	1102	1097	15.0	11.6	17.5	-	-	0.1
18	α -terpineol	1192	1189	0.1	tr	0.2	-	-	-
19	octanol acetate	1214	1214	-	0.1	tr	-	-	-
20	neral	1242	1238	-	0.2	0.2	-	-	-
21	<i>trans</i> -sabinene hydrate acetate	1267	1256	0.1	0.1	0.1	-	-	-
22	linalyl acetate	1257	1257	56.8	42.7	64.6	-	-	-
23	geranial	1267	1267	-	0.2	-	-	-	-
24	δ -elemene	1339	1338	-	-	-	-	-	0.1
25	<i>trans</i> -carvyl acetate	1340	1342	-	-	-	-	-	0.1
26	α -terpinyl acetate	1352	1349	0.3	0.3	0.4	-	-	-
27	neryl acetate	1368	1362	0.3	0.4	0.6	-	-	-
28	geranyl acetate	1386	1381	-	0.2	0.2	-	0.7	0.7
29	β -caryophyllene	1418	1419	0.6	0.5	1.0	-	-	0.2
30	α - <i>trans</i> -bergamotene	1437	1435	0.6	0.5	1.0	-	-	-
31	γ -muurolene	1477	1480	-	tr	-	-	6.7	-
32	germacrene D	1481	1485	0.1	-	0.1	-	-	12.4
33	bicyclogermacrene	1500	1500	tr	tr	tr	-	0.7	1.4
34	germacrene A	1503	1509	-	-	-	-	-	0.2
35	β -bisabolene	1509	1506	0.8	0.7	1.3	-	-	-
36	δ -cadinene	1523	1523	-	-	-	-	-	0.2
37	germacrene B	1561	1561	-	-	-	-	-	0.5
38	germacrene D-4-ol	1576	1576	-	-	-	-	-	0.2
39	γ -eudesmol	1632	1632	-	-	-	-	-	0.1
40	hinesol	1638	1642	-	-	-	-	-	0.1

(Continues)

TABLE 2 (Continued)

N°	Compounds	Retention index		Citrus samples					
				BEOs			CEOs		
				<i>C bergamia</i> 'Castagnaro'	<i>C bergamia</i> 'Fantastico'	<i>C bergamia</i> 'Femminello'	<i>C x myrtifolia</i>	<i>C mitis</i>	<i>C japonica</i>
		LRI _{exp}	LRI _{lit}	Relative content (%)					
41	α -eudesmol	1654	1654	-	-	-	-	0.7	-
42	<i>n</i> -heptadecane	1700	1700	-	-	-	-	-	0.3
43	<i>n</i> -octadecane	1800	1800	-	-	-	-	-	0.5
Chemical groups (%)									
	Monoterpene hydrocarbons			22.9	42.5	9.9	97.1	84.3	79.5
	Oxygenated monoterpenes			72.7	55.6	83.6	-	0.7	1.6
	Total monoterpenes			95.5	98.1	93.6	97.1	85.0	81.1
	Sesquiterpene hydrocarbons			2.2	1.7	3.4	-	7.4	15.1
	Oxygenated sesquiterpenes			-	-	-	-	0.7	0.5
	Total sesquiterpenes			2.2	1.7	3.4	-	8.1	15.6
	Non terpenes			-	0.1	-	-	-	0.9
	Total identified (%)			97.7	99.9	97.1	97.1	93.0	97.6
	N. of identified compounds			21	27	24	2	12	29

Abbreviations: LRI_{exp}, experimental linear retention index on DB-5 column; LRI_{lit}, Linear retention index recovered from the literature.²³

^anot detected;

^btr, trace.

performed by Head Space (HS) technique and not by direct immersion (DI-SPME) of the juice as carried out in this study. One of these papers is related to calamondin juice,^{17,38} where limonene (responsible for the fragrance) together with α -terpineol were identified as the main constituents, in agreement with the composition of our sample.

No data are reported in the literature on the volatile composition of the other *Citrus* juices studied herein. This is therefore the first report on their aroma profile directly studied, without heating the juice which can modify the composition as in the essential oil (producing artifacts), to identify the aroma constituents that we can intake simply by drinking the investigated juices.

3.4 | Statistical analyses

The dendrogram obtained with the Cluster Analysis (Figure 1A) evidenced two groups of taxa (similarity value = 38.92). One group was *C japonica*, *C x mirtyfolia*, and *C mitis* (similarity value = 58.1), while the other included *C bergamia* 'Castagnaro,' 'Fantastico,' and 'Femminello' (similarity value = 49.04); except for *C bergamia* 'Fantastico' obtained by direct immersion which was phytochemically more similar to *C japonica* and *C x mirtyfolia* obtained by direct immersion. The results obtained with the CA were confirmed by

NMDS (Figure 1B) with a good stress index (value = 0.09) that confirms the reliability of the segregation of the taxa into two groups. Five chemical compounds showed a significant discriminative function (*P*-value < .01) between the clusters. Samples of the first group were characterized by high amounts of limonene (3) while the second group differed in its predominance of linalool (5) and linalyl acetate (7), showing very low values in the other taxa.

The most interesting result from the multivariate statistical analyses is that the first NMDS axis can play a systematic role segregating the different taxa regardless of the extraction method, while along the second NMDS axis all the taxa are positioned in the bidimensional space according to the extraction method. Furfural (17) and 4-terpineol (26) are most significantly and directly correlated with the second axis. These latter compounds were obtained in the highest amounts with the direct immersion juice method but in very low percentages in the squeezed EO.

3.5 | Antimycotic activity

The results on the antifungal activity of the EOs tested against *Saprolegnia parasitica* are reported in Table 4. *Saprolegnia* spp. organisms are aquatic oomycetes known as water molds which severely

TABLE 3 Volatile compounds identified into the *Citrus* juices and analyzed by Direct Immersion-Solid-Phase MicroExtraction (DI-SPME)

N°	Compounds	Retention index		<i>Citrus</i> samples					
				<i>C bergamia</i> 'Castagnaro'	<i>C bergamia</i> 'Fantastico'	<i>C bergamia</i> 'Femminello'	<i>C x mirtyfolia</i>	<i>C mitis</i>	<i>C japonica</i>
		LRI _{exp}	LRI _{lit}	Relative content (%)					
1	3-methyl-3-buten-1-ol	734	731	- ^a	-	0.5	-	-	-
2	ethyl acetate	616	806	-	0.9	0.5	1.2	2.5	1.4
3	butyl methyl ether	811	811	-	-	-	-	1.8	-
4	1-butyl acetate	811	811	-	-	0.5	-	-	-
5	furfural	847	836	0.3	1.9	3.1	1.0	11.6	8.9
6	α -thujene	931	930	-	0.1	-	0.3	-	-
7	tricyclene	934	927	-	-	0.4	-	-	-
8	α -pinene	940	939	0.2	0.5	-	-	-	-
9	3-methyl-2,5-furandione	948	949	-	-	-	-	17.8	-
10	5-methyl furfural	969	964	0.4	-	-	2.0	1.6	0.8
11	sabinene	978	975	0.4	1.4	0.5	-	-	-
12	β -pinene	981	979	-	-	-	1.8	-	-
13	6-methyl-5-hepten-2-one	990	986	0.7	-	0.3	-	-	-
14	3-octanol	991	991	-	-	3.8	-	-	-
15	myrcene	993	991	1.3	2.6	0.7	1.3	1.8	1.3
16	α -terpinene	1019	1017	-	0.3	0.3	-	-	-
17	<i>o</i> -cymene	1026	1026	-	-	0.7	-	-	-
18	<i>p</i> -cymene	1028	1025	0.4	-	-	0.2	-	-
19	limonene	1032	1029	26.5	65.0	30.1	77.0	26.1	72.9
20	2-heptyl acetate	1044	1044	-	-	-	-	1.8	-
21	phenylacetaldehyde	1047	1045 ^b	-	-	-	-	0.7	-
22	(<i>E</i>)- β -ocimene	1053	1050	-	-	0.2	-	-	-
23	dihydrotagetone	1053	1053	-	-	-	-	-	0.5
24	γ -terpinene	1062	1060	2.0	9.1	3.9	-	-	-
25	<i>n</i> -octanol	1068	1068	-	-	-	-	0.2	-
26	<i>cis</i> -linalool oxide	1087	1074	0.4	-	1.0	1.1	-	-
27	terpinolene	1090	1089	-	1.8	-	-	-	-
28	<i>para</i> -mentha-2,4(8)-diene	1090	1088	0.9	1.4	-	-	-	-
29	linalool	1102	1097	19.2	-	11.9	1.0	2.1	-
30	maltol	1108	1108	0.6	0.3	-	0.2	-	1.3
31	<i>exo</i> -fenchool	1116	1122	1.3	-	0.3	-	-	-
32	<i>trans-para</i> -mentha-2,8-dien-1-ol	1125	1123	0.5	-	0.8	-	-	-
33	<i>cis</i> - β -terpineol	1144	1144	0.2	-	1.8	-	-	-
34	camphor	1148	1146	-	-	0.4	-	-	-
35	borneol	1169	1169	1.1	-	-	-	-	-
36	<i>cis</i> -linalyl oxide (pyranoid)	1174	1174	-	-	0.2	-	-	-
37	4-terpineol	1180	1177	4.7	0.3	3.6	0.5	2.2	-
38	α -terpineol	1189	1189	-	0.4	4.5	1.8	7.4	-
39	γ -terpineol	1192	1199	18.9	0.2	0.2	1.1	1.4	-
40	dihydrocitronellol	1196	1196	-	-	-	-	3.5	-
41	octanol acetate	1214	1214	0.2	0.4	0.6	-	-	-
42	citronellol	1231	1226	0.9	-	0.8	-	-	-

(Continues)

TABLE 3 (Continued)

N°	Compounds	Retention index		Citrus samples					
				<i>C bergamia</i> 'Castagnaro'	<i>C bergamia</i> 'Fantastico'	<i>C bergamia</i> 'Femminello'	<i>C x mirtyfolia</i>	<i>C mitis</i>	<i>C japonica</i>
		LRI _{exp}	LRI _{lit}	Relative content (%)					
43	carvotanacetone	1251	1247	0.3	-	-	-	-	-
44	geraniol	1259	1253	3.3	-	-	-	-	-
45	linalyl acetate	1260	1275	-	3.7	7.7	3.4	-	-
46	perilla alcohol	1295	1295	-	-	0.1	-	-	-
47	α -terpinyl acetate	1352	1349	0.2	-	-	-	-	-
48	neryl acetate	1368	1362	0.4	-	0.5	-	-	-
49	geranyl acetate	1386	1381	0.3	0.3	0.6	0.8	-	0.8
50	decanol acetate	1411	1408 ^b	-	0.2	0.4	-	-	-
51	β -caryophyllene	1418	1419	-	0.4	-	-	-	-
52	<i>trans</i> - α -bergamotene	1437	1435	0.4	1.3	1.6	-	-	-
53	<i>allo</i> -aromadendrene	1461	1460	-	-	0.2	-	-	-
54	germacrene D	1481	1485	-	-	-	-	-	0.5
55	β -bisabolene	1509	1506	0.3	1.5	0.9	-	-	-
56	guaial	1597	1601	0.3	-	-	-	-	-
57	10-epi- γ -eudesmol	1627	1624	0.8	-	-	-	-	-
58	γ -eudesmol	1634	1632	0.3	-	-	-	-	-
59	α -muurolol	1649	1646	0.3	-	-	-	-	-
60	α -cadinol	1655	1654	-	-	-	-	0.2	-
61	selin-11-en-4- α -ol	1655	1660	-	-	0.4	-	-	-
62	valerianol	1658	1655 ^b	-	-	-	-	1.4	-
63	β -bisabolol	1672	1675	-	0.3	0.4	-	-	-
64	tetradecanol	1675	1673	-	-	-	-	-	-
65	epi- α -bisabolol	1685	1685	0.3	0.4	0.6	-	-	-
66	juniper camphor	1695	1692	0.9	-	-	-	-	-
67	ethyl tetradecanoate	1796	1796 ^b	1.8	-	-	-	-	-
68	nootkatone	1807	1807	-	0.6	1.1	4.1	-	-
69	cryptomeridiol	1814	1814	-	-	-	-	0.1	-
70	α -vetivone	1843	1843	1.9	-	-	1.0	-	-
71	cyclohexadecanolide	1935	1935	0.5	-	-	-	-	-
72	isopropyl hexadecanoate	2025	2025	0.3	-	-	-	-	-
Chemical groups (%)									
Monoterpene hydrocarbons				31.6	82.3	36.9	80.6	27.9	74.3
Oxygenated monoterpenes				51.6	4.9	34.5	9.8	16.6	1.3
Total monoterpenes				83.2	87.3	71.3	90.4	44.4	75.5
Sesquiterpene hydrocarbons				0.6	3.2	2.7	-	-	0.5
Oxygenated sesquiterpenes				4.7	1.3	2.4	5.1	1.7	-
Total sesquiterpenes				5.3	4.5	5.1	5.1	1.7	0.5
Non terpenes				4.9	3.6	9.7	4.4	38.0	12.3
Total identified (%)				93.4	95.4	86.0	99.9	84.1	88.4
N. of identified compounds				38	25	38	18	18	9

Abbreviations: LRI_{exp}, experimental linear retention index on DB-5 column; LRI_{lit}, Linear retention index from the literature.²³

^anot detected.

^bNIST chemistry web book and citation herein.

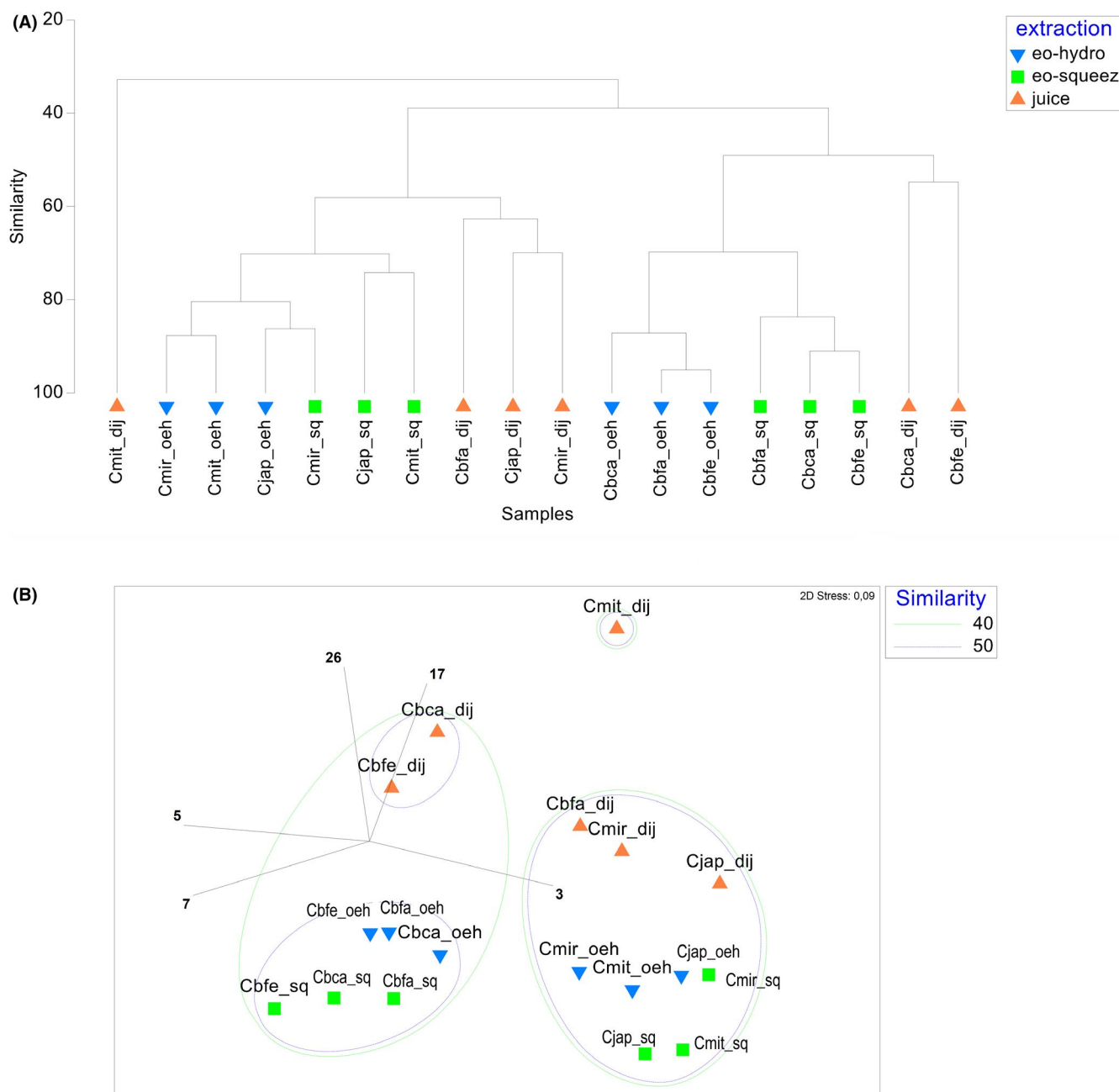


FIGURE 1 Multivariate analyses of the taxa x chemical compounds matrix. **A**, Dendrogram obtained by cluster analysis using average-linkage clustering (UPGMA). **B**, NMDS plot: the clusters obtained with CA were differently colored (in green taxa with a similarity of 40, in blue taxa with a similarity of 50). Taxa acronyms: Cbca, *C bergamia* 'Castagnaro'; Cbfa, *C bergamia* 'Fantastico'; Cbfe, *C bergamia* 'Femminello'; Cjap, *C japonica*; Cmir, *C × mirtyfolia*; Cmit, *C mitis*. Extraction methods: dij, direct immersion juice; eo-hydro, EO obtained by hydrodistillation; eo-squeez, EO obtained by squeezing. Abbreviations of chemical compounds: 3 = limonene, 5 = linalool, 7 = linalyl acetate, 17 = furfural, 26 = 4-terpineol. Compounds that showed a Pearson correlation > 0.8 with the ordination axes were reported. [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/fft.3603)]

compromise fish health in aquaculture, fish farms, and hobby fish tanks. Saprolegniosis occurs mainly in salmonid aquaculture but also in wild fish.³⁹ These oomycetes colonize the mature and immature stages of fish, as well as the eggs, causing hyphal growth showing as cotton-like mycelia on the body surface and gills. All the EOs from bergamot varieties and Kumquat showed good antimycotic activity, even though *C bergamia* 'Fantastico' exhibited the lowest MIC (minimum inhibitory concentrations).

The efficacy of CEOs for controlling *Saprolegnia* infections was verified by Nardoni and collaborators,²⁷ where commercial *C bergamia* EO showed an MIC value of 2.5%, together with *Cymbopogon flexuosum* and *Litsea cubeba* (MIC 2.5%), even though not at the high level of activity showed by *Thymus vulgaris* and *Origanum vulgare* EOs (MIC 0.5%). The relative low efficacy showed by BEO in the paper of Nardoni et al.²⁷ was attributed to the different composition of the bergamot EO (purchased from Flora Srl., Lorenzana-Pisa,

TABLE 4 Antifungal tests (MIC observed on the different strains, as percentages)

Essential oil tested	Fungal strains			
	<i>M canis</i>	<i>M gypseum</i>	<i>S parasitica</i>	<i>T mentagrophytes</i>
<i>Citrus bergamia</i> 'Fantastico'	0.5	0.5	0.5	0.5
<i>Citrus bergamia</i> 'Castagnaro'	5	5	0.75	5
<i>Citrus bergamia</i> 'Femminello'	5	>10	0.75	>10
<i>Citrus x myrtifolia</i>	>10	>10	>10	>10
<i>Citrus japonica</i>	0.5	0.5	2.5	0.5

Abbreviation: MIC, minimum inhibitory concentration.

Italy), with limonene 33.2%, linalool 14.6% and linalyl acetate 31.4%, γ -terpinene 6.4%, and β -pinene 5.4% as the main compounds, in comparison with the EO composition found herein after direct hydrodistillation of the fresh fruit, where the high percentage of linalool (34.3%) might be responsible for the good activity (see Table 1).

The antimycotic activity of CEOs (especially from the three cultivars of *C bergamia* and *C x myrtifolia*) in sweet almond oil solutions was also assayed against dermatophytes such as clinical isolates of *M canis*, *T mentagrophytes*, and *M gypseum*, respectively, which are responsible for human and animal infections. The results on the antifungal activity are reported in Table 4. *Citrus bergamia* 'Fantastico' showed the best antimycotic activity against all the tested dermatophytes (MIC 0.5%). This evidence was already reported in the study of Sanguinetti and collaborators⁴⁰ where they confirmed the high in vitro activity of bergamot oil (especially the distilled BEO) against a wide number of clinical isolates of several pathogenic dermatophytes, among these *T mentagrophytes*, *M canis*, and *M gypseum*. Our results far exceeded that observed in the paper by Nardoni et al⁴¹ where the tested *C bergamia* EO, purchased from Flora Srl., (Lorenzana-Pisa, Italy), had different compositions especially related to the percentages of the main constituents (limonene 27.1%, linalool 16.8%, and linalyl acetate 34.5%). Thus, the EO composition is one of the main requirements that must be known and considered before attributing a biological activity to a natural phytocomplex.

Very good activity was also showed by kumquat EO, with a MIC value of 0.5% against the tested dermatophytes. There are no previous reports in the literature on the antifungal activity of *C japonica*. Only *F crassulacea* EO showed remarkable activity against *Candida albicans* along with a strong antimicrobial effect.⁴² The antifungal activity of CEOs was reviewed by Jing and collaborators⁴³ especially against *Aspergillus* spp. (a common food contaminant), *Penicillium* genus (postharvest pathogens), *Alternaria alternata* and *Botrytis cinerea*, but the activity showed by CEOs from minor *Citrus* fruits was not revised. Unfortunately, it was not possible to evaluate the antimycotic activity of the CEOs obtained by cold pressing due to their very low amount.

4 | CONCLUSIONS

Minor *Citrus* fruits cultivated in Italy were studied for their EO and volatile juice composition. Limonene was the most abundant

compound in CEOs from the minor fruits, while BEOs changed its composition according to the different extraction method used and the bergamot varieties. This paper is the first report on the characterization of the juice volatilome from minor *Citrus* fruits by direct immersion as well as on the assessment of the antimycotic activity of *C japonica* essential oils against dermatophytic species. Moreover, CEO obtained by hydrodistillation of *C bergamia* 'Fantastico' was the most active in the antifungal tests, even though it was the less productive in the essential oil yield. Kumquat EO also showed a high effectiveness, especially against dermatophytes. Based on the present results, the use of these products would appear an interesting alternative in preparing topical herbal products for the treatment of affected animals or for environmental fungal control purposes.

CONFLICT OF INTEREST

The authors have no conflicts of interest to report.

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